Xiangyu Xie
中国科学技术大学
化学物理系，2026 届

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教育背景

合肥，中国 **中国科学技术大学** **2022 年秋 – 预计 2026 年春**

- **化学物理理学学士**，2026 届
- **GPA：** 3.23/4.3 – 成绩呈持续上升趋势：大一（2.6）、大二（3.4）、大三（3.5）
- **导师：**江俊教授（计算化学与 AI for Science）
- **相关课程：**量子力学、统计力学、数学物理方程、高等量子力学、量子化学、线性代数

科研经历

本科毕业论文 **导师：江俊教授** **2025 年秋 – 2026 年春**

基于等变图神经网络的室温磷光分子性质预测

- **数据集与预测目标：**面向约 6,000 个有机室温磷光候选分子，构建基于三维分子图的数据处理流程，并引入 $\Delta E_{S_1-T_1}$、$\mathrm{SOC}_{T_1\to S_1}$、$\mathrm{SOC}_{T_1\to S_0}$ 与 f_{osc,S_1} 等量子化学标签
- **模型架构：**设计并实现 8 层 EGNN_Sparse_Network，使用 21 维原子节点特征、6 维键边特征、三维坐标与等变消息传递建模分子几何信息
- **基线与训练：**在相同的 4,800/1,050/150 训练/验证/测试划分下，与字符级 SMILES Transformer 基线比较；仅在 EGNN 训练阶段使用旋转/平移数据增强
- **结果：**在测试集上，$\Delta E_{S_1-T_1}$、$\mathrm{SOC}_{T_1\to S_1}$、$\mathrm{SOC}_{T_1\to S_0}$ 与 f_{osc,S_1} 的 R^2 分别达到 0.8781、0.6263、0.9855 和 0.7844
- **结论：**表明显式三维几何信息与等变归纳偏置能够显著提升孤立分子 RTP 激发态性质预测能力，相比序列模型更适合该类任务
- **技能：** PyTorch、图神经网络、SE(3) 等变性、量子化学、Python、Linux

本科生科研 **导师：崔林松教授** **2023 年秋 – 2024 年秋**

基于双重态发光机制的发光材料设计与合成

- **项目：**参与本科生科研计划，完成课题研究并通过结题考核
- **分子设计与合成：**设计 TTM-AnQ 自由基，将 TTM 自由基与 9,10-蒽醌受体偶联；通过 Friedel–Crafts、Suzuki 偶联与氧化反应完成多步有机合成
- **光谱表征：**开展 UV-Vis、荧光、循环伏安（CV）与瞬态荧光测试；观察到受体引入后发射峰由 580 nm 红移至 695 nm
- **关键发现：**验证 SOMO 能级调控（$-2.82\,\mathrm{eV} \to -3.18\,\mathrm{eV}$），提出区别于传统给体功能化的发射调控思路，并完成受体功能化自由基发光体的探索性研究

本科生竞赛 **数智化赛道** **2024 年秋 – 2025 年春**

多尺度分子模拟驱动的物理化学实验教学创新

- **框架：**构建“实验 + 计算”一体化方案，将分子动力学模拟（GROMACS）与 DFT 计算（Gaussian 16, B3LYP/6-31G+g(d,p)）相结合
- **模拟：**构建正丁醇/环己烷体系（17:400 与 1:400），在 30°C 下完成能量最小化、平衡与生产模拟，并抽取 100 帧轨迹用于 DFT 计算
- **结果：**得到 1.78 D 的预测偶极矩（相对实验值 1.66 D 偏差 6.0%），并揭示无限稀释外推法的微观来源
- **奖项：**获第五届全国大学生化学实验创新设计大赛（数智化赛道）**华中赛区一等奖**

研究演示 **2026 年春至今**

MultiSpec-GeoDiff：光谱驱动的分子结构反演

- **项目链接：** GitHub
- **项目：**基于 200 条实验 NIST 红外光谱，构建光谱驱动分子结构反演的 Stage-I 可复现验证闭环，引入坐标感知光谱编码与谱距离偏置注意力机制，完成从光谱输入、候选检索到结果可视化的端到端流程

- **检索与重排序：** 实现候选分子检索、前向光谱重排序、Top-K 结果可视化、可追溯日志输出与 CI 测试流程，形成可运行、可复现的研究 demo

奖项与荣誉

- **中国科学技术大学优秀学生奖学金** – 因学业表现优秀获评（2024、2025）
- **中国科学技术大学强基计划专项补助** – 面向前 20% 学生（2025）
- **本科生竞赛华中赛区一等奖** – 第五届全国大学生化学实验创新设计大赛，数智化赛道（2025）

学术交流

科研访问 **多个科研院所**

- **2023 年 8 月：** 中国科学院大连化学物理研究所、中国科学院长春应用化学研究所、中国科学院金属研究所（沈阳）
- **2025 年 8 月：** 中国科学院上海有机化学研究所、中国科学院上海硅酸盐研究所、中国科学院苏州纳米技术与纳米仿生研究所
- **2025 年 7 月：** 香港大学、香港科技大学、香港中文大学、香港城市大学

助教经历

课程助教

- **2024 年秋：** 复变函数（化学类）– 组织习题讨论课、批改作业并提供一对一辅导
- **2025 年春：** 数学物理方程（B）– 组织解题训练并编写补充材料
- **2026 年春：** 量子物理 – 展现扎实的理论基础与课程理解

技术技能

编程与工具

- **机器学习 / 几何深度学习：** PyTorch、图神经网络、EGNN、张量场网络、SE(3)-Transformer、Transformer 架构
- **编程与 AI 辅助开发：** Python；Claude Code、Cursor、Codex、OpenCode；AI 辅助编程、智能体式编码与科研原型快速开发
- **计算化学：** DFT/TD-DFT 计算、量子化学软件（Gaussian、VASP）、分子可视化
- **数学基础：** 线性代数、群论、概率统计、复变函数
- **通用工具：** LaTeX、Git、Linux/Unix 命令行、数据分析（numpy、pandas、matplotlib）

个人优势

- **扎实的数学基础：** 熟悉理论化学与数学物理核心原理，能够支撑计算化学与 AI4S 建模工作
- **分析与科研执行能力：** 能够提出研究问题、构建分析路径，并推进复杂科研项目从构想到完成
- **交叉与协作能力：** 兼具湿实验合成与计算建模经验，具有跨课题组、跨机构合作与交流经历
- **第一性原理思维与 AI4S 建模：** 能够从物理与数学基础出发分析科学问题，并将几何深度学习、等变建模与 Transformer 架构应用于 AI for Science 场景

University of Science and Technology of China
Department of Chemical Physics, Class of 2026

XIANGYU XIE

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EDUCATION

Hefei, China | **University of Science and Technology of China** | **Fall 2022 – Expected Spring 2026**

- **B.S. in Chemical Physics**, Class of 2026
- **GPA:** 3.23/4.3 – Demonstrating consistent upward trajectory: Freshman (2.6), Sophomore (3.4), Junior (3.5)
- **Advisor:** Prof. Jun Jiang (Computational Chemistry & AI for Science)
- **Relevant Coursework:** Quantum Mechanics, Statistical Mechanics, Mathematical Physics Equations, Advanced Quantum Mechanics, Quantum Chemistry, Linear Algebra

RESEARCH EXPERIENCE

Undergraduate Thesis | **Supervisor: Prof. Jun Jiang** | **Fall 2025 – Spring 2026**

Prediction of Room-Temperature Phosphorescence Molecular Properties Based on Equivariant Graph Neural Networks

- **Dataset & Targets:** Built a 3D molecular graph pipeline for about 6,000 organic RTP candidate molecules with quantum-chemistry labels for $\Delta E_{S_1-T_1}$, $\mathrm{SOC}_{T_1\to S_1}$, $\mathrm{SOC}_{T_1\to S_0}$, and f_{osc,S_1}
- **Model Architecture:** Designed and implemented an 8-layer EGNN_Sparse_Network using 21-dim atom node features, 6-dim bond edge features, 3D coordinates, and equivariant message passing
- **Baseline & Training:** Compared against a character-level SMILES Transformer baseline under the same 4,800/1,050/150 train/validation/test split; applied rotation/translation augmentation only during EGNN training
- **Results:** Achieved test-set R^2 values of 0.8781, 0.6263, 0.9855, and 0.7844 for $\Delta E_{S_1-T_1}$, $\mathrm{SOC}_{T_1\to S_1}$, $\mathrm{SOC}_{T_1\to S_0}$, and f_{osc,S_1}, respectively
- **Conclusion:** Showed that explicit 3D geometry and equivariant inductive bias improved isolated-molecule RTP excited-state property prediction over the sequence baseline
- **Skills:** PyTorch, Graph Neural Networks, SE(3) Equivariance, Quantum Chemistry, Python, Linux

Undergraduate Research | **Supervisor: Prof. Linsong Cui** | **Fall 2023 – Fall 2024**

Design and Synthesis of Light-Emitting Materials Based on Doublet-Emissive Mechanism

- **Project:** Undergraduate Research Program; successfully completed and passed final evaluation
- **Molecular Design & Synthesis:** Designed TTM-AnQ radical by coupling TTM radical with 9,10-anthraquinone acceptor; executed multi-step synthesis via Friedel–Crafts, Suzuki coupling, and oxidation
- **Characterization:** Conducted UV-Vis, fluorescence, CV, and transient fluorescence measurements; observed emission red-shift from 580nm to 695nm upon acceptor incorporation
- **Key Finding:** Demonstrated SOMO energy modulation (-2.82eV → -3.18eV), suggesting an emission-tuning pathway beyond traditional donor-functionalization; conducted an exploratory undergraduate study of acceptor-functionalized radical emitters

Undergraduate Competition | **Digital Intelligence Track** | **Fall 2024 – Spring 2025**

Multiscale Molecular Simulation-Driven Physical Chemistry Experiment Teaching Innovation

- **Framework:** Developed "Experiment + Computation" integrated protocol combining MD simulations (GROMACS) with DFT calculations (Gaussian 16, B3LYP/6-31G+g(d,p))
- **Simulations:** Built n-butanol/cyclohexane models (17:400 and 1:400); performed energy minimization, equilibrium, and production runs at 30°C; extracted 100 trajectory frames for DFT computation
- **Results:** Achieved 1.78D predicted dipole moment (6.0% deviation from experimental 1.66D); revealed microscopic origin of infinite dilution extrapolation method

- **Award: First Prize, Central China Division** of the 5th National Undergraduate Chemistry Experiment Innovation Design Competition (Digital Intelligence Track)

Research Demo **Spring 2026 – Now**
MultiSpec-GeoDiff: Spectra-Driven Molecular Structure Inversion **GitHub:** GitHub
- **Project:** Built a runnable Stage-I feasibility loop for spectra-driven molecular structure inversion using 200 experimental NIST IR spectra with coordinate-aware spectral encoding and spectral-distance-bias attention
- **Retrieval & Reranking:** Developed candidate retrieval, forward-spectrum reranking, Top-K visualization, traceable outputs, and CI-tested reproducible demo workflow

Awards and Honors

- **USTC Outstanding Student Scholarship** – Awarded for academic excellence (2024, 2025)
- **USTC Strengthening Basic Disciplines Plan Grant** – Top 20% of students eligible (2025)
- **Undergraduate Competition, Central China Division – First Prize** – 5th National Undergraduate Chemistry Experiment Innovation Design Competition, Digital Intelligence Track (2025)

Academic Exchanges

Research Visits **Multiple Institutes**
- **August 2023:** Dalian Institute of Chemical Physics, Changchun Institute of Applied Chemistry, Shenyang Institute of Metal Research
- **August 2025:** Shanghai Institute of Organic Chemistry, Shanghai Institute of Ceramics, Suzhou Institute of Nano-Tech and Nano-Bionics
- **July 2025:** Hong Kong University, HKUST, Chinese University of Hong Kong, City University of Hong Kong

Teaching Experience

Teaching Assistant
- **Fall 2024:** Complex Functions (for Chemistry) – Led discussion sections, graded assignments, provided one-on-one tutoring
- **Spring 2025:** Mathematical Physics Equations (B) – Facilitated problem-solving sessions, developed supplementary materials
- **Spring 2026:** Quantum Physics – Demonstrating strong command of theoretical foundations

Technical Skills

Programming and Tools
- **Machine Learning / Geometric Deep Learning:** PyTorch, Graph Neural Networks, EGNN, Tensor Field Network, SE(3)-Transformer, Transformer architectures
- **Programming & AI-assisted Development:** Python; Claude Code, Cursor, Codex, OpenCode; agentic coding and research-oriented prototyping workflows
- **Computational Chemistry:** DFT/TD-DFT calculations, Quantum Chemistry Software (Gaussian, VASP), Molecular Visualization
- **Mathematical Foundations:** Linear Algebra, Group Theory, Probability and Statistics, Complex Functions
- **General Tools:** LaTeX, Git, Linux/Unix command line, Data Analysis (numpy, pandas, matplotlib)

Strengths

- **Solid Mathematical Foundation:** Strong grasp of theoretical chemistry and mathematical physics principles
- **Analytical & Professional Competence:** Derive novel research questions, formulate rigorous analytical approaches, and execute complex research projects from conception to completion
- **Interdisciplinary & Collaborative:** Combine wet lab synthesis with computational modeling; experience working across multiple research groups and institutions
- **First-principles reasoning and AI4S modeling:** Analyze scientific problems from physical and mathematical foundations; understand geometric, equivariant, and Transformer-based deep learning models for AI for Science.